Technical Deployment & Operations Runbook

Document Reference: CAW-TECH-RUN-2026-01
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System
Organisation: Citizens Advice Wandsworth (CAW)
Target Audience: DevOps Engineers, Cloud Infrastructure Administrators, and IT Leads
Target Environment: Pilot (europe-west2 London)
Status: Approved Runbook
Classification: Internal / Technical

0. Canonical Names

Every name below has been verified against the live Google Cloud estate. Earlier revisions of this runbook, the configuration reference and the custom domain guide each used a different invented project name (caw-case-ace-prod-2026, caw-case-ace-pilot-london and advice-intelligence-prod respectively). None of those projects exists. This table is now the single source of truth and any document that disagrees with it is wrong.

1. System Architecture as Deployed

Case Ace v2.0 is deployed as a static single page application served by Firebase Hosting, plus a stateless Express/Node.js backend running on Cloud Run. There is no virtual machine, no Docker Compose stack, no Nginx instance and no database. The SPA does the recording, identifier detection and redaction entirely in the browser; the backend exists only to authenticate advisers and to mint short-lived, downscoped Google Cloud credentials.

```
flowchart LR
    A["Adviser Browser (Chrome / Edge)"] -->|"HTTPS / TLS 1.3"| B["Firebase Hosting: case-ace-app"]
    A -->|"HTTPS / TLS 1.3"| C["Firebase Hosting: case-ace-api"]
    C -->|"rewrite"| D["Cloud Run: case-ace-api (europe-west2)"]
    D -->|"STS downscoped token"| E["Google Cloud STT v2 (europe-west2)"]
    D -->|"STS downscoped token"| F["Vertex AI Gemini (europe-west2)"]
    A <-->|"SRTP / WSS"| G["Cisco Webex Telephony Cloud"]
```

Infrastructure requirements

Frontend hosting: Firebase Hosting, two sites, both on the case-ace-v2 project.

Backend runtime: Cloud Run, europe-west2, gen2 execution environment.

Container build: Cloud Build. There is no requirement for Docker on any workstation.

Cloud infrastructure: the case-ace-v2 Google Cloud project with billing enabled.

Identity provider: Microsoft Entra ID for adviser SSO and RBAC, with TOTP as the interim login route while the tenant is unavailable (see section 5).

[!IMPORTANT]
Residency caveat. Cloud Run compute, Speech-to-Text and Vertex AI are all pinned to europe-west2. Firebase Hosting, however, is a global anycast content delivery network, so TLS terminates at whichever Google edge location is nearest the adviser, which may be outside the United Kingdom. Request bodies are not retained at the edge, and no client audio is ever sent to the backend, but the claim “all traffic terminates in the UK” is not currently accurate and should not be made in the DPIA in that form. Moving the API to a regional external Application Load Balancer in europe-west2 would make the claim hold end to end. This is an open decision, not a settled position.

2. Environment Configuration & Secret Management

Cloud Run holds configuration as service environment variables. There is no .env file on any server, and no service account key file anywhere: the service authenticates as its attached service account, which is the keyless pattern Google recommends and which avoids a long-lived credential sitting on disk.

[!WARNING]

backend/src/config/index.ts contains a development fallback value for JWT_SECRET which is committed to the repository and therefore public to anyone with repository access. If the service is deployed without JWT_SECRET set, adviser session tokens are signed with a key that is not secret, and a third party could mint a valid session. JWT_SECRET must be supplied from Secret Manager on every deployment.

Create the secret once:

```
# Generate a 64-byte random secret and store it in Secret Manager
openssl rand -base64 64 | tr -d '\n' | \
  gcloud secrets create case-ace-jwt-secret \
    --project=case-ace-v2 \
    --replication-policy=user-managed \
    --locations=europe-west2 \
    --data-file=-

# Allow the runtime service account to read it
gcloud secrets add-iam-policy-binding case-ace-jwt-secret \
  --project=case-ace-v2 \
  --member="serviceAccount:case-ace-api-sa@case-ace-v2.iam.gserviceaccount.com" \
  --role="roles/secretmanager.secretAccessor"
```

---

3. Step-by-Step Deployment

Step 1: Provision the runtime service account

The service currently runs as the project's default Compute Engine service account, which carries the broad Editor role. That is more privilege than this service needs and should be replaced with the dedicated, least-privilege account below before the pilot opens.

```
gcloud config set project case-ace-v2
gcloud config set compute/region europe-west2

gcloud iam service-accounts create case-ace-api-sa \
  --display-name="Case Ace v2 Backend Runtime Service Account"

# Least privilege: Speech-to-Text and Vertex AI only, plus token minting for downscoping
gcloud projects add-iam-policy-binding case-ace-v2 \
  --member="serviceAccount:case-ace-api-sa@case-ace-v2.iam.gserviceaccount.com" \
  --role="roles/speech.client"

gcloud projects add-iam-policy-binding case-ace-v2 \
  --member="serviceAccount:case-ace-api-sa@case-ace-v2.iam.gserviceaccount.com" \
  --role="roles/aiplatform.user"

gcloud projects add-iam-policy-binding case-ace-v2 \
  --member="serviceAccount:case-ace-api-sa@case-ace-v2.iam.gserviceaccount.com" \
  --role="roles/iam.serviceAccountTokenCreator"
```

No service account key is created or downloaded. Cloud Run supplies the identity to the container at runtime.

Step 2: Build and deploy the backend

Cloud Build performs the container build inside Google Cloud from the repository source, so no local Docker installation is required. infrastructure/gcp/cloudbuild.yaml is the build definition and infrastructure/docker/Dockerfile.backend is the image definition.

[!NOTE]
An organisation policy on adviceintelligence.tech constrains resource locations, so the default Cloud Build staging bucket in the US is rejected with
'us' violates constraint 'constraints/gcp.resourceLocations'. The
--gcs-source-staging-dir flag below points at a europe-west2 bucket and is mandatory.

[!IMPORTANT]
The backend is not compiled to JavaScript. backend/tsconfig.json sets
"noEmit": true alongside "allowImportingTsExtensions": true, because the codebase imports modules with explicit .ts extensions and TypeScript will not emit output in that mode. npm run build in the backend workspace is therefore a type check that produces no dist directory, and the container runs the TypeScript sources directly using Node's built-in type stripping. Any Dockerfile that expects a dist directory will fail with
COPY failed: stat app/dist: file does not exist.

```
# Build the image
gcloud builds submit \
  --project=case-ace-v2 \
  --region=europe-west2 \
  --config=infrastructure/gcp/cloudbuild.yaml \
  --gcs-source-staging-dir=gs://case-ace-v2-build-staging/source

# Deploy it
gcloud run deploy case-ace-api \
  --project=case-ace-v2 \
  --region=europe-west2 \
  --image=europe-west2-docker.pkg.dev/case-ace-v2/case-ace/backend:2.0.0 \
  --service-account=case-ace-api-sa@case-ace-v2.iam.gserviceaccount.com \
  --allow-unauthenticated \
  --ingress=all \
  --min-instances=1 --max-instances=10 \
  --cpu=1 --memory=512Mi --port=8080 \
  --set-env-vars=APP_ENV=pilot,GCP_REGION=europe-west2,GCP_PROJECT_ID=case-ace-
v2,ALLOW_TOTP_IN_PILOT=true,CORS_ORIGIN=https://caseace.adviceintelligence.tech \
  --set-secrets=JWT_SECRET=case-ace-jwt-secret:latest
```

--allow-unauthenticated and --ingress=all are required because Firebase Hosting reaches the service over its public Cloud Run endpoint. Authentication is enforced by the application, not by Cloud Run IAM.

Step 3: Build and deploy the SPA

```
# VITE_APP_ENV must be set. The environment resolver in client/src/config/environments.ts
# fails closed to 'local', so a build without it produces an SPA that points at
# http://localhost:8080 and silently does nothing useful in production.
VITE_APP_ENV=pilot npm run build

firebase deploy --only hosting --project case-ace-v2
```

firebase.json deploys both hosting targets: app serves client/dist with the SPA rewrite and the hardened header set, and api rewrites every path to the case-ace-api Cloud Run service.

Step 4: Connect the custom domains

DNS is already configured at IONOS. Both hostnames are CNAME records pointing at their Firebase Hosting sites:

Each hostname must also be registered against its site inside Firebase Hosting, or the site returns “Site Not Found” even though DNS resolves correctly. Add them in the Firebase console under Hosting !’ the relevant site !’ Add custom domain, or:

```
firebase hosting:sites:list --project case-ace-v2
```

[!NOTE]
Cloud Run domain mappings are not available in europe-west2. That is why the custom domains terminate at Firebase Hosting and are rewritten to Cloud Run, rather than being mapped directly to the Cloud Run service.

---

4. Post-Deployment Smoke Test

```
# 1. Backend health and region pinning
curl -s https://api.caseace.adviceintelligence.tech/health

# Expected:
# {"status":"healthy","environment":"pilot","gcpRegion":"europe-west2",
#  "isSyntheticOnly":false,"timestamp":"2026-..."}
#
# If this returns an HTML page headed "Congratulations | Cloud Run", the placeholder
# image is still deployed and Step 2 has not taken effect.

# 2. SPA reachability and security headers
curl -s -I https://caseace.adviceintelligence.tech | \
  grep -iE "^http/|content-security-policy|strict-transport-security"

# Expected: HTTP/2 200, a CSP beginning default-src 'none', and an HSTS header.
# "Site Not Found" means the custom domain is not yet connected to the case-ace-app site.

# 3. Ephemeral credential minting (requires a valid adviser session token)
curl -s -X POST https://api.caseace.adviceintelligence.tech/api/v1/credentials/issue \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <ADVISER_JWT>" \
```

```
  -d '{"purpose":"speech-to-text"}'
```

5. Authentication Route

The pilot goes live with TOTP as the active login route and Entra ID SSO configured but disabled, so SSO can be enabled later as a configuration change rather than a code change. This is a deliberate, documented override of the pilot's Entra-only default, recorded in docs/authentication-and-authorisation.md section 3.2 and implemented as the ALLOW_TOTP_IN_PILOT environment variable.

To switch to Entra ID SSO, remove ALLOW_TOTP_IN_PILOT from the Cloud Run service and redeploy. No rebuild of the SPA is required.

6. Rollback and Disaster Recovery

Cloud Run keeps every revision, so rollback is a traffic change rather than a rebuild.

```
# List revisions, newest first
gcloud run revisions list --service=case-ace-api \
  --project=case-ace-v2 --region=europe-west2

# Send all traffic back to a known good revision
gcloud run services update-traffic case-ace-api \
  --project=case-ace-v2 --region=europe-west2 \
  --to-revisions=<PREVIOUS_REVISION>=100
```

Firebase Hosting keeps previous releases and can be rolled back from the Hosting console release history, or by redeploying a known good build.

Advisers are not blocked by a rollback. Where the service is unavailable, the Business Continuity Procedure applies and advice continues with manual note taking in Casebook. No advice session is ever delayed by the unavailability of Case Ace.